An exact jet model with two packets

$F(x, y) = (x + xy,\ y + x^2)$ at $(0, 0)$. Coordinate packets in both input and output.

$DF(0) = I$: $\quad DF(0)e_x = (1, 0), \qquad DF(0)e_y = (0, 1).$

Output 1

H_1

	x	y
x	0	**1**
y	**1**	0

$D^2F_1(0)[e_x, e_y] = D^2F_1(0)[e_y, e_x] = 1$

Output 2

H_2

	x	y
x	**2**	0
y	0	0

$D^2F_2(0)[e_x, e_x] = 2$

$$D^2F(0)[h, k] = (h_x k_y + h_y k_x,\ 2h_x k_x)$$

$$\tfrac{1}{2}D^2F(0)[h, h] = (h_x h_y,\ h_x^2)$$

$F(h) = h + \frac{1}{2}D^2F(0)[h, h]$ exactly. All higher jets vanish.

Cells are ordered Hessian responses: the two mixed slots and the Taylor factor 1/2 are both retained.